

02.06.2021

**Chief Executive Officer,
For Subam Industries**

To
The Principal,
T.J.S. Engineering College,
Peruvoyal, Near Kavaraipettai,
Tamil Nadu 601206.

Dear Sir,

Sub: Request for involvement in collaborative research project with your
R&D Team -Reg.

Greeting from Subam Industries, Chennai

I hope this letter finds you in good health and high spirits. I am writing to propose collaboration between T.J.S Engineering College and Subam Industries for a project work initiative. This letter is regarding the 5 KW Roof Top Solar PV Implementation research grant with your R&D team in the college. Having thoroughly examined your college, we are eager to say that we would be interested in working with you on a study project. We are aware of your institution's standing and competence in the engineering and technology fields, and we think working together will provide important outcomes for both of our businesses.

We respectfully ask that you provide us an acceptance of project proposal details inorder to proceed with it. We are excited to contribute significantly to the field of Electrical and Electronics Engineering and collaborate closely with T.J.S. Engineering College.

I appreciate your consideration.

Yours Sincerely,
G. Karandhi
Chief Executive Officer,



[Signature]
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206



GST NO: 33AFZPK9630C1Z7

☎ 9444034474 / 9940143213 | ☎ 044-48534593

✉ subamkitchens@gmail.com / subameqep@gmail.com 🌐 www.subamkitchens.in / www.subamkitchens.com

No:14, Subam Campus, Sanjay Gandhi Nagar, Perumal Agaram, Thiruverkadu, Chennai – 600077, Tamilnadu, India.



T.J.S. ENGINEERING COLLEGE

(Approved by AICTE, Affiliated to Anna University, Chennai & Accredited by NAAC)
An ISO 9001:2015 Certified Institution



Dr. J. Prakash, M.E., Ph.D., MISTE,
Principal,
E-mail: principal@tjsec.in,
Ph: 044-27967602, 7373730006.

Date: 16.06.2021

To

The Chief Executive Officer,
Subam Industries (P) Ltd,
No.14, Perumalagaram Salai,
Sanjay Gandhi Nagar,
Chennai, Tamil Nadu-600077

Dear Sir,

Sub: Collaborative Research Opportunity with Subam Industries - Reg.

Thank you for offering R&D work to our institution, we are interest to establishing a research collaboration with your esteemed institution as per your requirement.

T.J.S. Engineering College, Peruvoyal, was established by Shri T.J.GOVINDARAJAN (FOUNDER & CHAIRMAN) in the year 2009. Our college is approved by AICTE, New Delhi and affiliated with Anna University, Chennai. We are offering five UG programs in Mechanical, Electrical and Electronics Engineering (EEE), Electronics and Communication Engineering (ECE), Computer Science Engineering (CSE) and Artificial Intelligent and Data Science (AI&DS) and Three PG programs in Master of Business Administration (MBA), Master of Very Large Scale Integration (VLSI), and Master of Computer Science Engineering. Our college is also involved in fostering research and consulting work in technical competence. Therefore, I am writing this letter to express our interest in establishing research collaboration with Subam Industries.

As stated above, we have potential faculty members from the core of Electrical and Electronics Engineering and they have expertise in the area of solar PV implementation and are also ready to submit a research proposal on 5 KW Roof Top Solar PV Implementation.

The purpose of this research collaboration is to combine our respective strengths, knowledge and resources to tackle specific challenges and capitalize on emerging opportunities.

Thank you for considering this opportunity. We look forward to hearing from you soon



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Yours Sincerely,

T.J.S.Nagar, Peruvoyal, Kavaraipeetai, Near Redhills, Gummidipoondi Taluk, Thiruvallur Dist - 601 206.
Phone : 044 27967602, 27967604, 27967605, Cell : 7373730006 / 9788877774

E-mail : tjsivanandam@gmail.com, tjsivanandam@gmail.com Website : www.tjsec.in

02.07.2021

**Chief Executive Officer,
for Subam Industries.**

To
The Principal,
T.J.S. Engineering College,
Peruvoyal, Near Kavaraipettai,
Tamil Nadu - 601206

Dear Sir,

Sub: Submission of Research Project proposal - Reg.

We are pleased to collaborate with TJS Engineering college for extending your technical support, with reference to the letter dated 16.06.2021, I am writing in response to your letter regarding the research grant offered in topic of solar PV implementation. We are excited to express our willingness to engage in a research project work with T.J.S. Engineering College.

We recognize the expertise and reputation of your institution in the field of Engineering and Technology, and we believe that collaboration between our organizations will yield significant results. So you are requested to send us a research proposal stating about the outlining, scope, methodology, timelines and also allocate the team members for the same.

We look forward to working closely with T.J.S. Engineering College and creating a lasting impact in the field of electrical and electronics engineering. The duration of the project work may be limited to 09 months from the project started. We would request you to submit the project proposal and all the other relative documents.

Yours Sincerely,

G. K. K. K.
Chief Executive Officer



J. S. A.
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206



GST NO: 33AFZPK9630C1Z7

☎ 9444034474 / 9940143213 | ☎ 044-48534593

✉ subamkitchens@gmail.com / subameqep@gmail.com 🌐 www.subamkitchens.in / www.subamkitchens.com

No:14, Subam Campus, Sanjay Gandhi Nagar, Perumal Agaram, Thiruverkadu, Chennai - 600077, Tamilnadu, India.



T.J.S. ENGINEERING COLLEGE

(Approved by AICTE, Affiliated to Anna University, Chennai & Accredited by NAAC)
An ISO 9001:2015 Certified Institution



Dr. J. Prakash, M.E., Ph.D., MISTE,
Principal,
E-mail: principal@tjsec.in,
Ph: 044-27967602, 7373730006.

Date: 12.07.2021

To

The Chief Executive Officer,
Subam Industries (P) Ltd,
No.14, Perumalagaram Salai,
Sanjay Gandhi Nagar,
Chennai, Tamil Nadu-600077

Dear Sir,

Sub: Submission of proposal and assigning of faculty members to do Collaborative research work with Subam Industries – Reg.

Greetings from the T.J.S. Engineering College (TJSEC)

With reference to the letter dated (02.07.2021), TJSEC is permitted the following faculty members to do collaborative research with Subam Industries and a proposal on the mentioned title "5KW ROOF TOP SOLAR PV IMPLEMENTATION" is submitted along with this letter. The following faculty members are assigned to do research work with Subam Industries.

List of Faculty members

S. No	Name of PI & Co-PI	Designation and Specialization	Contact information
1.	Dr. J. Prakash	Principal/ Principle investigator	9443275330 principal@tjsec.in
2.	Dr. I. Arul Doss Adaikalam	Associate Professor, Co-Investigator	9865773737 iadaikalam@gmail.com




PRINCIPAL

T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206

Yours Sincerely,


PRINCIPAL

T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206

T.J.S. Nagar, Peruvoyal, Kavaraipettai, Near Red Hills, Gummidipoondi Taluk, Thiruvallur Dist - 601 206.
Phone : 044 27967602, 27967604, 27967605, Cell : 7373730006 / 9788877774

E-mail : tjsivanandam@gmail.com, tjsivanandam@gmail.com Website : www.tjsec.in

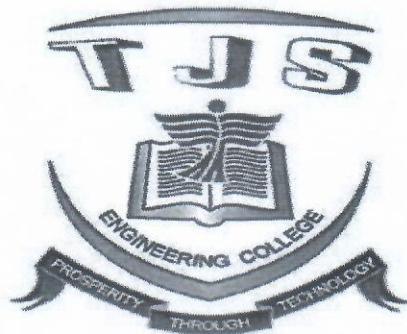
Submission of Research proposal
on
5KW ROOF TOP SOLAR PV IMPLEMENTATION

Submitted to Subam
Industries

Submitted by
Dr. J. Prakash,
Principal,
Principal Investigator

&

Dr. I. Arul Doss Adaikalam,
Associate Professor,
Co-Investigator




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

GENERAL INFORMATION ABOUT THE PROJECT AND THE PROJECT MEMBERS: -

1. Project Title: "5KW Roof Top Solar PV Implementation."
2. Name of the Institution and address: T.J.S. Engineering College,
Peruvoyal, Near Kavareipettai,
TamilNadu - 601 206.
3. Details of the Project Team:

PRINCIPAL INVESTIGATOR	
Name	Dr. J. Prakash
Date of Birth	20-JULY-1981
Highest Qualification	PhD
Designation	Principal
Department	EEE
Institution/ University	T.J.S. Engineering College
Complete Address with Pin code	T.J.S. Engineering College, Peruvoyal, Near Kavareipettai, Puduvoyal, TamilNadu-601 206.
Mobile number	9443275330
Email ID	principal@tjsec.in

CO-INVESTIAGTOR	
Name	Arul Doss Adaikalam I
Date of Birth	23-MARCH-1984
Highest Qualification	PhD
Designation	Associate professor
Department	EEE
Institution/University	T.J.S. Engineering College
Complete Address with Pin code	T.J.S. Engineering College, Peruvoyal, Near Kavareipettai, Puduvoyal, TamilNadu-601 206.
Mobile Number	9865773737
Email-ID	iadaikalam@gmail.com

4. Duration (months): 9 months.
5. Total cost (in Rs): 4.76 lakhs




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavareipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

CURRICULUM VITAE (PRINCIPAL INVESTIGATOR)

	Dr. J. PRAKASH, M.E., Ph.D., MISTE S/o. C.Jayachandran Naidu Plot no: 7, Srinivasan Street, Extension VI, VGN Mahalakshmi Nagar, Thiruverkadu, Chennai, Tamilnadu, India -600077
Phone no: +91 6379599795	
+919443275330	
E-mail : prakash_ies@yahoo.co.in	
Professional Summary	
Experience	: 21 years (Teaching.)
Educational Details	
Ph.D.,	Electrical Engineering VIT University, Vellore (2016)
M.E.	Energy Engineering Anna University, 2006 (First Class, 7.544 CGPA) (College of Engineering, Guindy)
B.E.	Electrical and Electronics Engineering University of Madras, 2002 (First Class, 70%) (Pallavan engineering college, Kanchipuram)

Work Experience (Total 19 years Teaching Experience)

Principal	T.J.S. Engineering College
September 16, 2021 to till date.	Peruvoyal, Kavaraipeitai, Chennai.

Principal	Jaya College of Engineering and Technology
January 24, 2020 to September 15, 2021	Poonamalle, Chennai

Professor, HOD/EEE and Vice Principal	Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College,
June 15, 2018 to January 24, 2020	Avadi, Chennai

Associate Professor	R.M.K College of Engineering and Technology,
(June 08, 2011 to May 31, 2018)	Chennai
	Associate Professor (June 2012 to May 31, 2018) Assistant Professor (June 2011 to May 2012)

Senior lecturer	Sri Ramachandra University, Porur
(May 14, 2009 to June 07, 2011)	Chennai.

Assistant professor	Lord Venkateshwara Engineering College
(Nov 2006 to May 13, 2009)	Walajabadh, Kancheepuram
	Assistant professor (July 2007 to May 2009)



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeitai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

Lecturer (May 2004 - Oct 2006)	P. T. Lee Chengalvaraya naicker college of Engg and Tech kancheepuram
--	--

Lecturer (June 2002 – May 2004)	Meenakshi College of Engineering, Uthiramerur.
---	---

Additional Responsibilities

- HOD at the Department of Electrical Engineering and Vice Principal at Vel Tech High Tech Dr Rangarajan Dr Sakunthala Engineering College (VTHT).
- I was involved as QS I GAUGE Survey and NBA Coordinator of the college (VTHT) and also involved in autonomous process.
- As part of NBA preparation for RMKCET, I am responsible for the preparatory efforts during SAR report preparation and ensuring that the necessary infrastructure and documents are in place.
- During NAAC visit to RMKCET, I was primarily responsible for the hospitality of the visiting experts on both days of inspection.
- As YRC (Youth Red Cross) Coordinator of the college (RMKCET), I organize various blood donation camps and social welfare activities at the surrounding villages. Also, I am involved in organizing tree plantation camps in coordination with top notch IT companies such as Infosys (Corporate Social Responsibilities activities).
- I am involved in taking care of the transport infrastructure of the college (RMKCET).
- I was involved as AICTE CII Survey Coordinator of our college and also involved in consultancy work.
- I have been actively involved in organizing International Conference at College level and various technical symposiums at the Department level.
- Acted as Placement Coordinator for the Department of EEE at RMKCET
- Acted as Organizing Committee Member for the one day Conclave of all Self Financing Engineering Colleges of Tamil Nadu on 20.05.2017.
- HOD – EEE and Executive Director at Lord Venkateshwara Engineering College
- Primarily responsible for the establishment of Lord Ayyapa Institute of Engineering and Technology during the year 2009 and Sri Ramachandra institute of technology during the academic year 2011.



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraiyil,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Patents

1. Title of the invention : **ENHANCEMENT OF SAFETY MEASURES IN METRO RAIL USING PLC AND SCADA**

Application No. 201641018081 A

Publication Date : 11/11/2016

2. Title of the invention : **ELECTRIC POWER DISTRIBUTION EMPLOYING ENSEMBLE MACHINE LEARNING BASED CYBER-PHYSICAL SYSTEM**

Application No. 202241006399 A

Publication Date : 25/02/2022

Academic Achievements

LIST OF PUBLICATIONS Scopus Author ID: 57205242403

International Journals

- **J.Prakash**, S. Harshavardhan Naidu, Izzatdin Abdul Aziz, Jafreezal Jaafar Reward-based residential wireless sensor optimization approach for appliance monitoring, **Soft Computing**, Springer Publisher, Issue 10/2021.
- **J.Prakash**, K.Gunalan, KM.Sundaram. Design and Development of Solar Photovoltaic System Using Single-Phase MLI, **Innovations in Electrical and Electronics Engineering**, Pg.639-651(Scopus indexed)
- KM Sundaram, D Nageswari, **J Prakash**. (2022) Cuckoo search assisted fuzzy logic algorithm for smart WSN routing system. **International Journal of Ad Hoc and Ubiquitous**, Vol. 40, pg.106.Inderscience Enterprises Ltd. Publisher.
- Iraj Sadegh Amiri, **J. Prakash**, M.Balasaraswathi, V.Sivasankaran, T.V. P. Sundararajan (2019) DABPR: a large-scale internet of things-based data aggregation back pressure routing for disaster management, **Wireless Networks**, **springer**, <https://doi.org/10.1007/s11276-019-02122-3>, (SCI & Scopus indexed)
- P.Muruganantham, **Prakash.J**, S,Vidhyasagar, C.S.Boopathi (2019). Design of Polarization Splitter using Elliptically Dual Core – Cladding Photonic Crystal Fiber **Results in Physics**, Vol.13, (SCI & Scopus indexed)
- S.Jayaprakash, K.Gunalan, **J.Prakash**. Topology and Performance Analysis of Cascadable Nine Level Inverter by Packed UCell Invertor and using Multi-Mode Synchronized PWM Schemes, **Journal of Green Engineering (JGE)** Vol.10. issue.9.
- U. Vinay Sahoo K. R. Sugavanam, A. Mutharasan, **J. Prakash**, P. Gowtham, M. Ramprasanth, MPPT in Partially Shaded PV System with the Use of WODE Technique, **Advances in Greener Energy Technologies**

- **Prakash J**, Mr. Sugavanam K R, Mr. Srikrishnakumar.S, Kokila S, Abin K, Taraipettai,



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Periyar Kavaraipttai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Hamshini. R. (2019) Design And Development Of Control Scheme For Solar PV System Using Single Phase Multilevel Inverter **Emerging Trends in Computing and Expert Technology, Lecture Notes on Data engineering and communications technologies** pg.1556-1569.

- Mr.Senthil Kumar R , Mr. Sri Krishnakumar S , **Dr. Prakash J** , Deepak Raju S Aravindsamy A and Prakash R, (2019) Polymers Based Material as A Safety Suit for High Power Utilities Working, **Emerging Trends in Computing and Expert Technology, Lecture Notes on Data engineering and communications technologies** pg.1211-1217.
- **Prakash, J., S.K. Sahoo and K.R.Suguvanam** (2016). Design of Coordinated Control Scheme for Hybrid Resonant Boost Converter and Multi Level Inverter, **Indian Journal of Science and Technology**, Vol.9, No.11, pp.1-11. (Scopus indexed)
- **Prakash, J., S.K. Sahoo, S. Prabhakar Karthikeyan, I. Jacob Raglend** (2015). Design of PSO-Fuzzy MPPT Controller for Photovoltaic Application, Power Electronics and Renewable Energy Systems, **Lecture Notes in Electrical Engineering** (Springer India)Vol. 326, pp. 1339-1348. (Scopus indexed)
- **Prakash, J., S.K. Sahoo** (2014). Design of Soft Switching Interleaved Boost converter for Photovoltaic application, **Research Journal of Applied Sciences, Engineering and Technology**. Maxwell Scientific Organization, Vol. 9 No .4, p.296. (Scopus indexed)
- **Prakash, J., S.K. Sahoo, S.P. Karthikeyan** (2012). A novel technique for common mode- voltage Elimination and DC- Link Balancing in Three-Level Inverter, **international Review on modeling and simulations (IREMOS)**, Vol.5, No.2, pp.840- 845.(Scopus indexed)

International conference

- **Prakash.J. Gunalan.K, Mohanasundaram.K** (2019) Design and Development of Solar Photo Voltaic System using Single Phase MLI, 4th International Conference on [Electrical and Electronics Engineering, springer ICIEEE2019 held on 26.07.2019 &27.09.2019 at Guru Nanak Institutions, Hyderabad. (Scopus indexed)
- **Prakash J, Mr. Sugavanam K R, Mr. Srikrishnakumar.S, Kokila S, Abarna T, Hamshini R.** (2019) Design And Development Of Control Scheme For Solar PV System Using Single Phase Multilevel Inverter, “2nd International Conference on Emerging Current Trends in Computing and Expert Technology”(COMET 2K19)” on March 22nd and 23rd, 2019 at Panimalar Engineering College, Chennai
- **Mr.Senthil Kumar R , Mr. Sri Krishnakumar S , Dr. Prakash J , Deepak Raju S Aravindsamy A and Prakash R,** (2019) Polymers Based Material as A Safety Suit for High Power Utilities Working, “2nd International Conference on Emerging Current Trends in Computing and Expert Technology”(COMET 2K19)” on March 22nd and 23rd, 2019 at Panimalar Engineering College, Chennai
- **Prakash, J., Saisidhartha. H, Raghavel. S. P, Shanmugasundaram. A** (2017). Coordinated control scheme in solar PV fed Boost converter and hybrid multilevel inverter in the International Conference on international intellectual convergence on advances in science and engineering (IIASE-2017) held on 03.03.2017, 04.03.2017.at Aalim Muhammed Salegh College of Engineering, Chennai.



[Handwritten signature]
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeetai,
 Gummidipoondi Taluk,
 Tiruvallur Dist - 601 206.

- **Prakash, J., S.K. Sahoo (2013).** Modelling of Photovoltaic System and Design of MPPT controller using PSO Algorithm International Conference on Environment & Energy 2013 held on 16.12.13-17.12.13.at The **Galadari Hotel, Colombo, Sri Lanka.**
- **Prakash, J., S.K. Sahoo, S. Prabhakar Karthikeyan, I. Jacob Raglend (2014).** Design of PSO- Fuzzy MPPT Controller for Photovoltaic Application at International Conference on Artificial Intelligence and Evolutionary Algorithms in Engineering Systems ICAEES-2014 organized by Noorul Islam Centre for Higher Education, Kumaracoil. (Scopus indexed)
- **Prakash, J., S.K. Sahoo, S. Prabhakar Karthikeyan. (2012).** Sinusoidal Output Voltage H-bridge Multilevel Inverters, IET Chennai 3rd International Conference on Sustainable Energy and Intelligent Systems (SEISCON 2012), p. 264. (Scopus indexed)
- Sahoo, S.K., Ramulu, A. **Prakash, J.** Performance analysis and simulation of five level and seven level single phase multilevel inverters IET Chennai 3rd International Conference on Sustainable Energy and Intelligent Systems (SEISCON 2012), Volume 2012, Issue 624 CP, 2012, Pages 285-291. (Scopus indexed)
- **Prakash, J., S.K. Sahoo (2012).** A Review on power electronics converters for solar photovoltaic system, 5th International Conference, on Science, Engineering and Technology (SET) organized by VIT University, Vellore, Tamilnadu, India.

National Conference

- Design and Fabrication of Battery based tricycle (Electric Vehicle) for physically challenged person by using power electronic device, Proceedings of National conference on recent trends in Power systems and drives, SSN College of Engineering, Chennai, April 2006, pp 66.

Paper presentation

(National and state level symposium)

Number of participation : 09

Number of Prizes awarded : 04

Papers: WAP {wireless application protocol}

Neuro – Fuzzy controller

Teaching Interests



- Circuit analysis
- Electrical Engineering
- Renewable energy sources
- Power Electronics
- Microprocessors
- Electric drives and control
- Electrical machines
- Embedded system

[Handwritten signature]

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeetai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 008

Academic Project

B.E.,

- ☐ **TITLE** : Design of Three Phase Amorphous met glass core Distribution Transformer
- CLIENT** : Indo Tech Transformers Ltd, Chennai.
- DESCRIPTION** : The main aim of our project is to reduce 70% of core loss by replace amorphous core instead of CRGO

M.E.,

- ☐ **TITLE** : Design, Development and Fabrication of Battery based tri cycle for Physically challenged person by using power electronics drives.
- DESCRIPTION** : The main aim of our project is to implement clean transportation instead of emission transportation by using modern electric propulsion system.

Ph.D

- ☐ **TITLE** : Design of Direct Control PV System with Single Phase Multilevel Inverter
- DESCRIPTION** : The most important goal of this research, it consist of dc-dc converter and Single Phase Multilevel Inverter to solve problems in the supply of quality power. The design helps to reduce the size of passive components, reduce ripple in output voltage and input current, reduce THD level, improve impedance matching, maximize the output power, improve the efficiency.

Honours/Awards

Member, Board of Studies (Engineering), Sri Ramachandra University, Porur,
Chennai- 600 116

Computer Skills

- ☐ **Languages** : C, C++, Embedded C,
- ☐ **Operating systems** : RTOS(VX WORKS, MICROCOSII)
- ☐ **Database** : Ms – Access
- ☐ **RDBMS** : Oracle 8

Conference/workshop Organized

- ☐ New energy sources from chemical and bio-chemical fields, AICTE_ISTE, March 10-23, 2004
- ☒ Third National Conference on "Recent Trends in Power Engineering, Power Electronics and Control Systems" (NCRTPPC-2014) on 24h March 2014.
- ☐ Fourth National Conference on "Recent Trends in Power Engineering, Power Electronics and Control Systems" (NCRTPPC-2015) on 23rd February 2015.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Training Program Organized

- ☐ Organized a series of training programs on embedded systems for Practicing Engineers
- ☐ Organized three training programs on RTOS Software
- ☐ Mr. J. Prakash, Associate Professor was the resource person for "Importance of Professional bodies" for the "Two Days Refresher Course on Academic Excellence" on

16.06.14 & 17.06.14 conducted by RMK College of Engineering and Technology.

Workshop Participated

Number of participation : 20

Facilities Established

1. Established Electrical Machines and Power System Simulation Laboratory in the Department of Electrical and Electronics Engineering, Lord Venkateswara Engg. College, kancheepuram, India,
2. Established a new engineering college in the name of Lord Ayyappa Institute of Engineering and Technology, Uthukadu, Kancheepuram dist.(2009)

Personal Details

Date of Birth	: 20/07/1981
Father name	: C. Jayachandran
Marital Status	: Married
No. of Children	: Two
Contact Address	: 7, Srinivasan street, VGN Mahalakshmi Nagar, Perumalagaram, Thiruverkadu, Chennai Tamil Nadu, India, Pincode:-600077
Language known	: Tamil, Telugu, English.
Contact No.	: 9443275330 : 6379599795

Declaration

The above details furnished by me are true to the best of my knowledge.



Date:
(Dr.J.Prakash)


PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Reference:

1. Dr.G.Prabhakaran
M.E.,Ph.D., Vice
Principal and
Ex.Professor, NIT
Tiruchy
SRM Institute of Science and
Technology, Ramapuram, Chennai,
Tamilnadu, India
Phone: +919840747717
2. Mr.M.Shanmugasundar
am. M.E., Associate
Professor/CSE (Rtd.)
R.M.K.College of Engineering and
Technology Chennai, Tamilnadu,
India Phone:+919790972638



PRINCIPAL

T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

CURRICULUM VITAE (CO-INVESTIGATOR)

Staff Name : ARUL DOSS ADAIKALAM.I

Date of Birth : 23.MARCH.1984

Age: 34

Contact Ph No : 9865773737

Address : NO 7, DHANDAPANI NADAR STREET,
PONNERI - 601204



Educational Qualification : M.E

Teaching / Non-Teaching : TEACHING

Date of Joining : 26.06.2015

Present Designation : ASSISTANT PROFESSOR

Experience : 9 Years 7 Months

S.No	Organization Name	Designation	Year of Working
1	T.J.S ENGG COLLEGE	Associate Professor	2015-TILL DATE
2	SARDAR RAJA COLLEGE OF ENGINEERING	Assistant Professor	2009 -15
3	P.S.N INSTITUTE OF TECHNOLOGY AND SCIENCE	Assistant Professor	2009 - 09

Training Undergone:

S.No	Training Details	Date of Training
1.	One day Pre-Conference Tutorial on "Smart Grid Systems"	(26/08/2011)
2.	One day Seminar on "Opportunities and Challenges in Renewable Power Generation"	(10/02/2012)
3.	Two days Seminar on "Research Methodologies and Issues in Electrical Engineering"	(8&9/06/2012)
4.	Two days FDP on "Smart Grid Technologies – Issues and Challenges"	(18&19/11/2013)
5.	One day Short-Term Course on "Recent Trends in Smart Grid Technologies"	(26/04/2014)
6.	Attended a FDTP Program on Power system Operation and control	9 to 20/12/2016
7.	Presented a Paper in the NATIONAL CONFERENCE in RVS Padhmavathy College of Engineering and Technology	30 & 31/03/2017
8.	Attended a FDTP Program on Protection and Switchgear	22 to 29/05/2017
9.	Presented three Paper in the NATIONAL CONFERENCE in Velammal Institute of Technology	14&15/06/2017



PRINCIPAL
T.J.S ENGINEERING COLLEGE
Peruvoyal, Kavarai, Tamil Nadu,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

10.	Presented a Paper in the INTENATIONAL CONFERENCE in Jai Shriram Engineering College	07/03/2019
11.	Presented a Paper in the INTERNATIONAL CONFERENCE ON EMERGING TRENDS IN COMPUTING, ELECTRICAL AND ELECTRONICS - 2019	17/03/2019

Paper published so far:

S.No	Name of Paper	Year
1.	Demand response program with different elasticities, I Arul Doss Adaikalam, CK Babulal. Iranian Journal of Science and Technology, Transactions of Electrical	2020
2.	Wireless Charging Station System for Autonomous Drone. GI Loretta, SJ Rubavathy, IAD Adaikalam, K Malarkodi, R Ganesan, 2023 4th International Conference on Smart Electronics and Communication(ICOSEC)	2023
3.	Closed Loop Performance Analysis of AFZ Converter with Improved Voltage Stability. SS Merlin, IAD Adaikalam, D Kirubakaran, R Ganesan, SJ Rubavathy, 2023 4th International Conference on Smart Electronics and Communication(ICOSEC)	2023

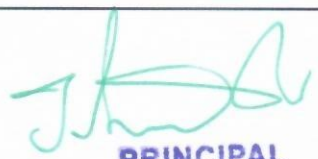



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipettai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

SHORT DESCRIPTION ABOUT THE PROPOSAL

<u>Project Name:</u>	5KW ROOF TOP SOLAR PV IMPLEMENTATION
<u>Abstract</u>	In The world's sunniest regions are found in India. With 300–325 bright days per year, the majority of India receives 4.7 kWh of solar energy per square meter each day. India has an abundance of solar resources; a location receives over 5,000 trillion kWh from 3000 hours of sunshine annually. India has no trouble using solar energy. Today By providing various economic and financial incentives, the government promotes the production of power from a variety of renewable energy sources, including wind, solar, small hydro, and biomass. Aside from this, state governments are purchasing electricity at a discounted rate from renewable energy projects. The nation now has 29,536MW of installed renewable energy capacity, including 19,933MW from wind, 2079MW from solar, 3746 MW from small hydro, and 3776 MW from bio energy. For the decentralized production of power, the Ministry of New and Renewable Energy offers a variety of renewable energy technologies. 10,752 communities have been powered thus far utilizing different renewable energy technologies. There have been installed 138 MW of decentralized solar power plants, 2.55 lakhs solar street lights, 9.93 lakhs solar house lighting systems, 9.39 lakhs solar lanterns, and 9.93 lakhs solar home lighting systems. For Solar PV based Power Plants, Ministry of New and Renewable Energy has announced a variety of financial incentives, including lowered customs charges, excise duty exemptions, and expedited depreciation. Local policies that include buyback, wheeling, and banking of the generated electricity by State Electricity Boards, among other incentives, are used to promote solar power generation at the state level.
<u>Estimated Budget</u>	The total Estimated Budget will be about 4,76,000/- for Design and Implementation of 5kW Roof Top Solar PV Implementation.
<u>Beneficiaries</u>	Subam Industries, Chennai.
<u>Project Duration</u>	09 month




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavarai, Tamil Nadu,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

LIST OF ANNEXURES :

Annexure Numbers	Description
I	Design Basis
II	Technical Specification
III	Scheduled of Exclusions
IV	Estimation

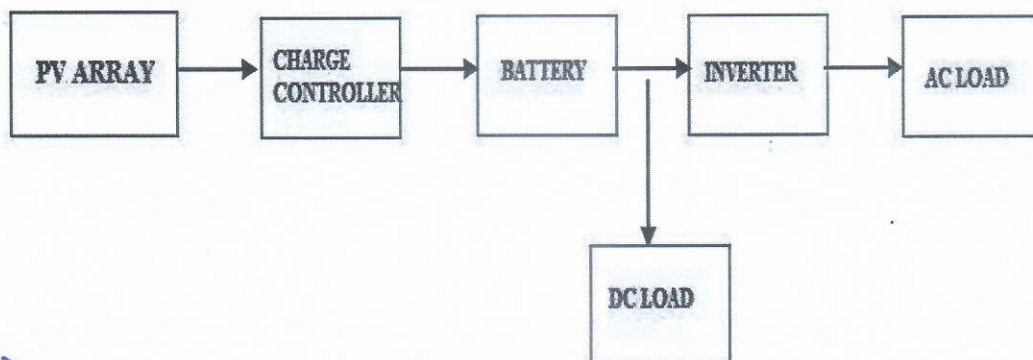



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

ANNEXURE - I DESIGN BASIS

Basic System Description

Solar Photovoltaic power generator consists of solar modules in series and parallel connections, these convert solar radiations into DC electrical power at the pre-determined range of Voltages when ever sufficient solar radiation is available. The individual crystalline solar cells are connected together in a module (in series connection), which are hermetically sealed to survive in rugged weather conditions and ensures optimum performance during its ling life.



[Signature]
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipondi Taluk,
Thiruvallur Dist - 601 206.

ANNEXURE - II TECHNICAL SPECIFICATION

This chapter discusses in detail the technical aspects of the components that shall be used in the Power Plant. As indicated the following are the major components of the Power Plant.

S. No.	Item	Description
1	Solar Module	440Wp Mono Crystalline
2	Module Mounting	Fixed Tilt Angle
3	Inverter	String inverter for 5 kVA
4	Combiner box	Array Junction Boxes
5	Control Panels	LT(Low Tension)Panel
6	Cables	PVC Copper Cables
7	Lightning	Lightning Protection Units




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeitai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

ANNEXURE – III SCHEDULE OF EXECUTION

S.No	Nature of Work	Time line for the week
1.	Approval of Project	5 weeks
2.	Site assessments	6 weeks
3.	Design and engineering new solar panel system	8 weeks
4.	Acquiring of solar panel	4 weeks
5.	Installation of solar panel	5 weeks
6.	Final inspection and commissioning	8 weeks
7.	Completion of project report	2 weeks
TOTAL		38 weeks



A handwritten signature in green ink, appearing to be "Jhsa".

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

ANNEXURE – IV ESTIMATION

BILL OF MATERIALS

S.No	Description	Quantity	Unit	Price/Rs
1	Solar module (550Wp)	10	Sets	2,00,000
2	String Inverter (5kVA, 48V)	1	Nos	90,000
3	Lithium Battery (5kW, 48W)	1	Nos	12,000
4	Solar Installation Kit	For 5 Kwp	Set	70,000
5	Solar 5kW Net Metering	1	Set	20,000
6	Lightning Arrestor, Lightning cable with socket & clamp	1	Set	15,000
7	Earthing kit	1	Set	13,000
8	Labor contract with Raw materials			56,000
TOTAL				4, 76,000

Payment Terms	
Term – I	50 % of Total grant, after approval of design as an advance for Purchasing of Raw materials.
Term - II	25 % For Manufacturing process
Term - III	25% at the time of Delivery of proposed Project.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

10.09.2021

**Chief Executive Officer,
Subam Industries.**

To
The Principal,
T.J.S. Engineering College,
Peruvoyal, Near Kavareipettai,
Tamil Nadu 601206.

Dear Sir,

Sub: Approval of Research Proposal "5kW roof top Solar PV implementation" - Reg.

Based on the earlier discussion, I am writing to inform you that the research proposal submitted by T.J.S Engineering College for collaboration with Subam Industries has been reviewed and approved by the appropriate authorities at our company. We are pleased to inform you that the proposal on "5kW roof top Solar PV implementation" is approved, and the proposed amount of Rs 4, 76,000/- is sanctioned for the same.

The grant has start date of June 2021 to February 2022, Please note that completion of the project, Subam Industries will require the core team of your college to complete the project within the speculated time period.

Yours Sincerely,

G. Kalanithi
Chief Executive Officer,



J. J. A.
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavareipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

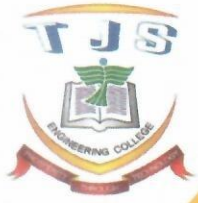


GST NO: 33AFZPK9630C1Z7

☎ 9444034474 / 9940143213 | ☎ 044-48534593

✉ subamkitchens@gmail.com / subameqep@gmail.com 🌐 www.subamkitchens.in / www.subamkitchens.com

No:14, Subam Campus, Sanjay Gandhi Nagar, Perumal Agaram, Thiruverkadu, Chennai – 600077, Tamilnadu, India.



T.J.S. ENGINEERING COLLEGE

(Approved by AICTE, Affiliated to Anna University, Chennai & Accredited by NAAC)
An ISO 9001:2015 Certified Institution



Dr. J. Prakash, M.E., Ph.D., MISTE,
Principal,
E-mail: principal@tjsec.in,
Ph: 044-27967602, 7373730006.

Date: 16.02.2022

UTILIZATION CERTIFICATE

I Dr. J. Prakash, hereby submit this Utilization Certificate to acknowledge the proper utilization of the grant amount of Rs.4,76,000 (Rupees Four Lakh Seventy Six thousand only) provided by Subam Industries. The grant was awarded for the research work entitled "5kW Roof Top Solar PV implementation" conducted at T.J.S. Engineering during the period from June 2021 to February 2022.

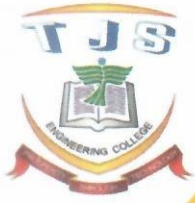
I further certify that the entire grant amount has been utilized judiciously and solely for the purpose stated in the research proposal.



PRINCIPAL/ PRINCIPAL INVESTIGATOR

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.



T.J.S. ENGINEERING COLLEGE

(Approved by AICTE, Affiliated to Anna University, Chennai & Accredited by NAAC)
An ISO 9001:2015 Certified Institution



Dr. J. Prakash, M.E., Ph.D., MISTE,
Principal,
E-mail: principal@tjsec.in,
Ph: 044-27967602, 7373730006.

Date: 16.02.2022

To

The Chief Executive Officer,
Subam Industries (P) Ltd,
No.14, Perumalagaram Salai,
Sanjay Gandhi Nagar,
Chennai, Tamil Nadu-600077

Dear Sir,

Sub: Completion of Project on 5kW Roof Top Solar PV
implementation - Reg.

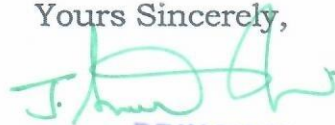
I am writing to inform you that the project entitled "5kW roof top Solar PV implementation" has been successfully completed by our team at T.J.S. Engineering College. Over the course of nine months, our dedicated team of faculty members led by Dr. J. Prakash worked diligently to ensure the smooth progression and timely completion of the project.

Furthermore, we would like to express our gratitude for the opportunity to collaborate with T.J.S Engineering College on this significant project. The trust and support you placed in our team allowed us to deliver a successful outcome. We also appreciate the cooperation and coordination on your end throughout the project's lifecycle.

Thank you for your trust and collaboration.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Yours Sincerely,

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Project Report
on
5KW ROOF TOP SOLAR PV IMPLEMENTATION

Submitted to Subam

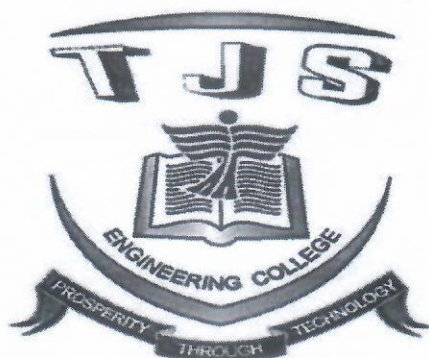
Industries

Submitted by

Dr. J. Prakash,
Principal,
Principal Investigator

&

Dr. I. Arul Doss Adaikalam,
Associate Professor,
Co-Investigator



J. Prakash

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 203.

5KW ROOF TOP SOLAR PV IMPLEMENTATION



A handwritten signature in green ink, appearing to be "J. J. S.", written over the printed name of the principal.

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Table of Contents :

Sl. No	Particulars
I	Abstract
II	Introduction
III	Technical Details
IV	Details of the Power plant
V	Project Implementation
VI	Conclusion
VII	Reference




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

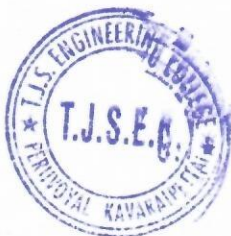
ABSTRACT

In The world's sunniest regions are found in India. With 300–325 bright days per year, the majority of India receives 4.7 kWh of solar energy per square meter each day. India has an abundance of solar resources; a location receives over 5,000 trillion kWh from 3000 hours of sunshine annually. India has no trouble using solar energy. Today By providing various economic and financial incentives, the government promotes the production of power from a variety of renewable energy sources, including wind, solar, small hydro, and biomass. Aside from this, state governments are purchasing electricity at a discounted rate from renewable energy projects. The nation now has 29,536MW of installed renewable energy capacity, including 19,933MW from wind, 2079MW from solar, 3746 MW from small hydro, and 3776 MW from bio energy.

For the decentralized production of power, the Ministry of New and Renewable Energy offers a variety of renewable energy technologies.

10,752 communities have been powered thus far utilizing different renewable energy technologies. There have been installed 138 MW of decentralized solar power plants, 2.55 lakhs solar street lights, 9.93 lakhs solar house lighting systems, 9.39 lakhs solar lanterns, and 9.93 lakhs solar home lighting systems.

For Solar PV based Power Plants, Ministry of New and Renewable Energy has announced a variety of financial incentives, including lowered customs charges, excise duty exemptions, and expedited depreciation. Local policies that include buyback, wheeling, and banking of the generated electricity by State Electricity Boards, among other incentives, are used to promote solar power generation at the state level.



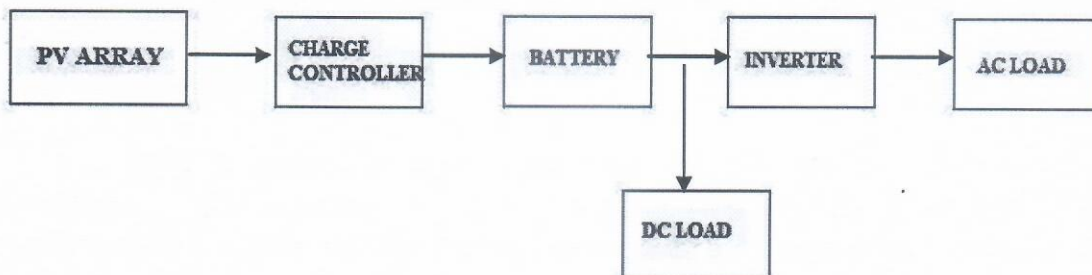
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

INTRODUCTION

Basic System Description

Solar Photovoltaic power generator consists of solar modules in series and parallel connections, these convert solar radiations into DC electrical power at the pre-determined range of Voltages when ever sufficient solar radiation is available. The individual crystalline solar cells are connected together in a module (in series connection), which are hermetically sealed to survive in rugged weather conditions and ensures optimum performance during its ling life.

In order to achieve a higher system voltage, modules are installed in arrow arrangement, called a string. A higher system voltage has the advantage of lesser installation work, higher efficiency of the entire plant and usage of smaller cross section cables. Calculated no. of strings is connected in parallel by cables in Junction Boxes. Outputs from many such junction boxes are connected in parallel in the Main Combiner Box (MCB). This Main Combiner Box output is fed to the central inverters/Power Control Unit (PCU) to invert solar generated DC power into conventional 3phase AC power. AC power from inverters will be fed to Low Voltage panel.



Operation Philosophy

Solar panels mounted in the field generate DC electric power. The DC electric power generated by the solar panels cannot be fed directly in to the utility grid. The inverters invert the direct current output from the solar array into grid compliant AC voltage, feeds it in to the utility grid system with proper protection and control. The grid connected inverter (GCI) range of inverters comes with built-in transformer that ensures galvanic isolation of the DC side from the AC network.

This is an important requirement for many utilities to permit connection of solar panels on to the grid. The system automatically starts up in the morning and begins to export power to the grid, provided there is sufficient solar energy and the grid voltage, frequency is within the range. If the grid goes out of range the inverter will be immediately disconnected and reconnected automatically at a pre-determined time after the grid comes back within range.



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Kallur Dist - 601 206.

TECHNICAL DETAILS

This chapter discusses in detail the technical aspects of the components that shall be used in the Power Plant. As indicated the following are the major components of the Power Plant.

S. No.	Item	Description
1	Solar Module	440Wp Mono Crystalline modules
2	Module Mounting Structure	Fixed Tilt Angle
3	Inverter	String inverter for 5 kVA
4	Combiner box	Array Junction Boxes
5	Control Panels	LT(Low Tension)Panel
6	Cables	PVC Copper Cables
7	Lightning	Lightning Protection Units

Solar PV Panel

Solar cells produce Direct Current (DC) electricity from light, which can be used to power equipment or to recharge a battery. Cells require protection from the environment and are usually packaged tightly behind a glass sheet. When more power is required than a single cell can deliver, cells are electrically connected together to form photo voltaic modules, or solar panels. A photo voltaic module is a packaged, interconnected assembly of photovoltaic cells, which converts sunlight into electrical power. The cells are hermitically sealed between glass and back cover (Tedlar) to protect them from harsh environments.

CRYSTALLINE MODULE

Technical Specifications for Crystalline Modules	
Capacity of PV Modules	250Wp/Mono
Output power– Pmax(Watts)	250Wp
Voltage at Pmax	30.16V
Current at Pmax	8.29A
Open-circuit voltage	37.88V
Short circuit current	8.75A
Module Efficiency(%)	15.2
Maximum system voltage(Volts)	IEC:DC1000V/UL:DC600V
Type of solar PV cell	Poly/Multi-Crystalline silicon
Solar cells	60 nos. of 6"siliconcells
Module output	MC4 terminations
Certification	IEC61215&ENIEC61730 Class A, Safety Class II



T.J.S. ENGINEERING COLLEGE
PERUVYOAL, KAVAYENPETA TALUK,
SRI GANESH DIST - 600 076.

Power warranty	25year limited warranty on power output &5-year product warranty 25-year performance warranty 10 years: 90% minimum performance. 25years: 80% minimum performance
----------------	--

MODULE MOUNTING STRUCTURE

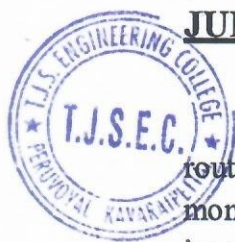
The module mounting structure is designed for holding suitable number of modules in series. The frames and leg assemblies of the array structures is made of MS hot dip galvanized of suitable sections of Angle, Channel, Tubes or any other sections conforming to IS:2062 for steel structure to meet the design criteria. All nuts & bolts considered for fastening modules with this structure are of very good quality of Stainless Steel. The array structure is designed in such a way that it will occupy minimum space without sacrificing the output from SPV panels at the same time it will withstand severe wind speed upto maximum 100 kmph.

Technical Specification – Module Mounting Structures	
Material	GI
Over all dimension	As per design
Wind rating	100km/hr
Tilt angle	15°- Fixed angle
Foundation	RCC/PCC As per design
Fixing type	SS304 fasteners

JUNCTION BOXES

In the Junction boxes, individual module strings are bundled and safely routed to the inverter. It is a combination of an exact, well-organized string monitoring system and a safety concept adapted to the PV technology. The junction boxes will have suitable cable entry points fitted with cable glands of appropriate sizes for both incoming and outgoing cables. They monitor the output of solar PV arrays. If difference between string outputs is too large, the operator is informed through monitoring system. Active disconnection allows string voltages to be measured separately.

These junction boxes are enclosed in an IP 65 rated stainless steel housing, making it ideal for long-term use in PV systems. In addition, the direct connection between the strings and the spring clamp connect or ensures a durable and safe installation.

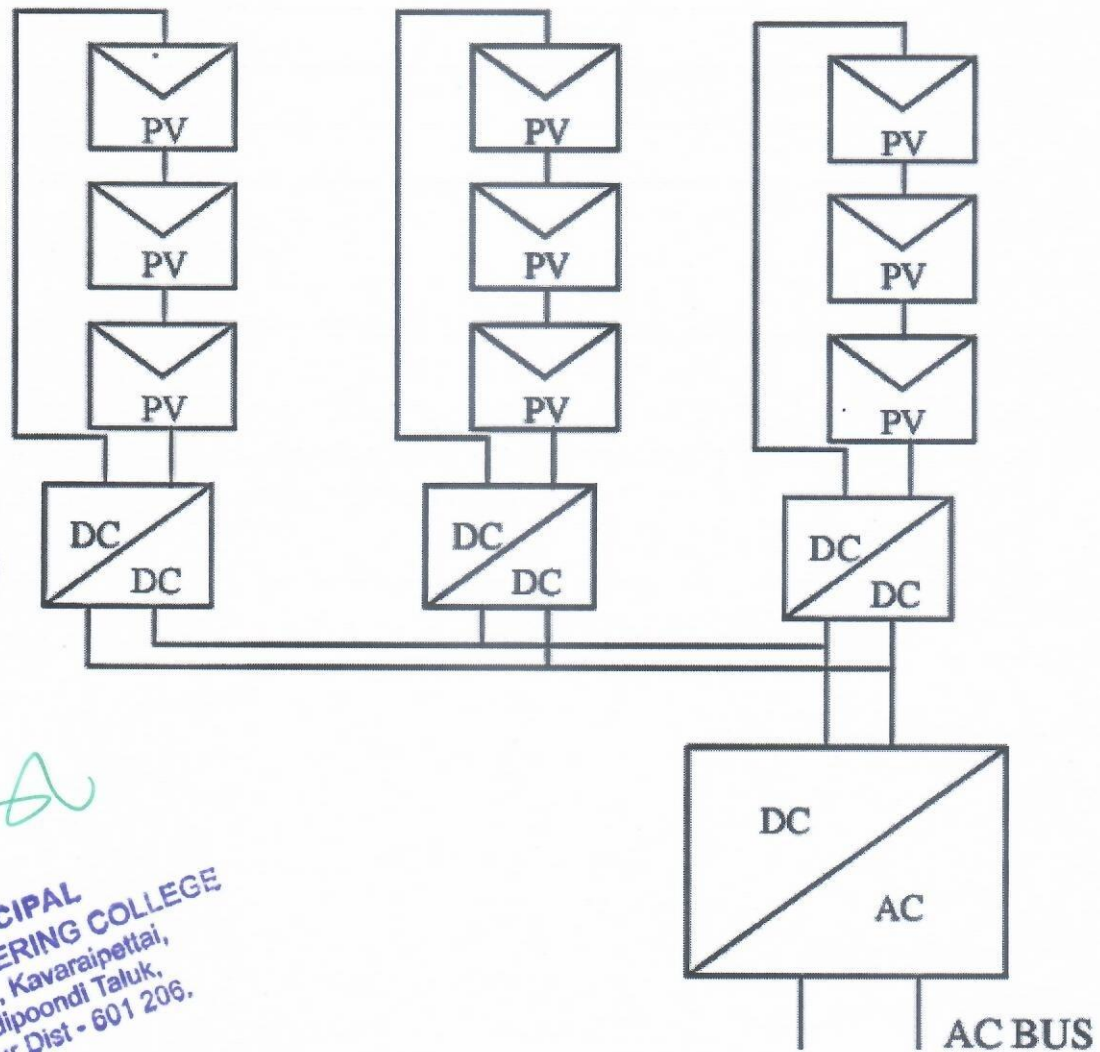


JBA
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvolai, Kavarai
Gummidipoondi Taluk,
Thiruvallur Dist. 601 204

Technical Specification – Combiner Boxes	
Material	Thermoplastic/Metal Sheet
Type	Dust, Vermin& Water proof
Hardware	SS304
Cable Gland	Thermoplastic
Protection	IP65

STRING INVERTER

A multi-function power conditioning system combining the functionality of a grid interactive solar inverter with a highly efficient conversion unit. The inverters achieve their distinguished efficiency factor through the use of IGBTs (Insulated Gate Bipolar Transistors) with Trench Gate Structure, and through the use of iron powder chokes and high quality transformers with losses less than 1%.



skd

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeetai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

String/Central inverters operate on MPPT (Maximum Power Point Tracking) mode to ensure maximum output from the solar generators at different ambient conditions. String inverters use higher system voltages to reach very high plant efficiency. Furthermore, installations can be expanded with additions of more modules without problems.

The design of the technical parameters has been optimized with regard to the operating time of the inverter. For example, the IGBTs not only increase the efficiency factor, but their high dielectric strength inhibits the transmission of voltage spikes. The intelligent minimum- performance cognition system protects the AC contactors and a fast over-current recognition system for all transistors protects the IGBTs.

With a wide range of input voltages from 350 to 1000 V, the inverter allows the largest possible range of module connection possibilities. For larger installations, the proposed inverters can be easily combined to provide higher performance.

CABLES

The size of the cables between array interconnections, array to junction boxes, junction boxes to PCU etc shall be so selected to keep the voltage drop and losses to the minimum. The bright annealed 99.97% pure bare copper conductors that offer low conductor resistance, they result in lower heating thereby increase in life and savings in power consumption. These wires are insulated with a special grade PVC compound formulated. The skin coloration offers high insulation resistance and long life.

Technical Specification – Cables	
Type	PV Insulated, sheath & UV resistant
Material	Copper
Voltage	Max.1100VforDC
Temperature	10–70°C
Colour	Red/Black/Yellow/Green

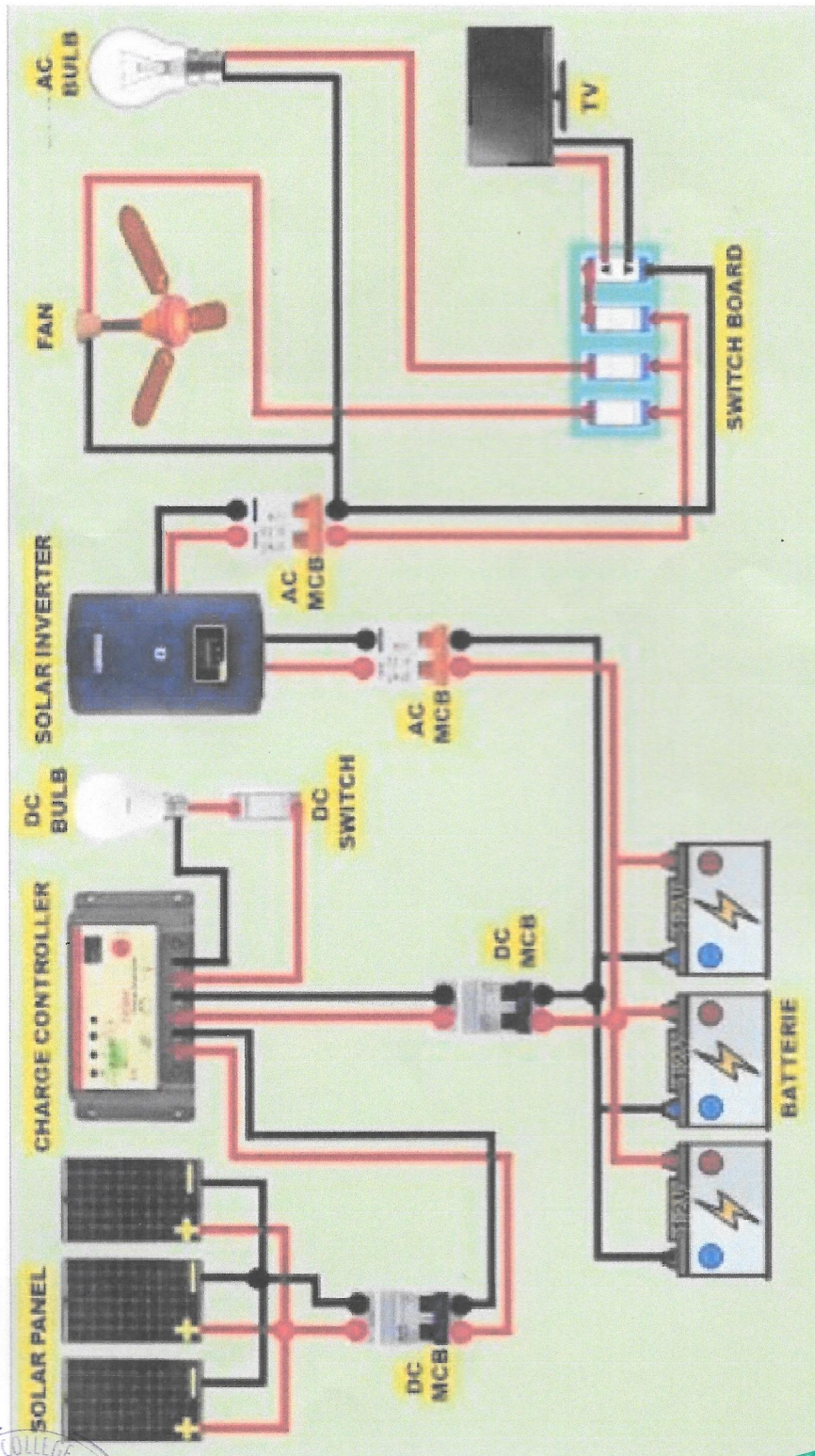
Cables are flexible & of annealed electrolytic grade copper conductor and shall confirm to IS1 554/694-1990 and are extremely robust and resist high mechanical load and abrasion.

Cable is of high temperature resistance and excellent weather proofing characteristics which provides along service life to the cables use d in large scale projects. The connectors /lugs of copper material with high current capacity and easy mode of assembly are proposed.



shd
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeetai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

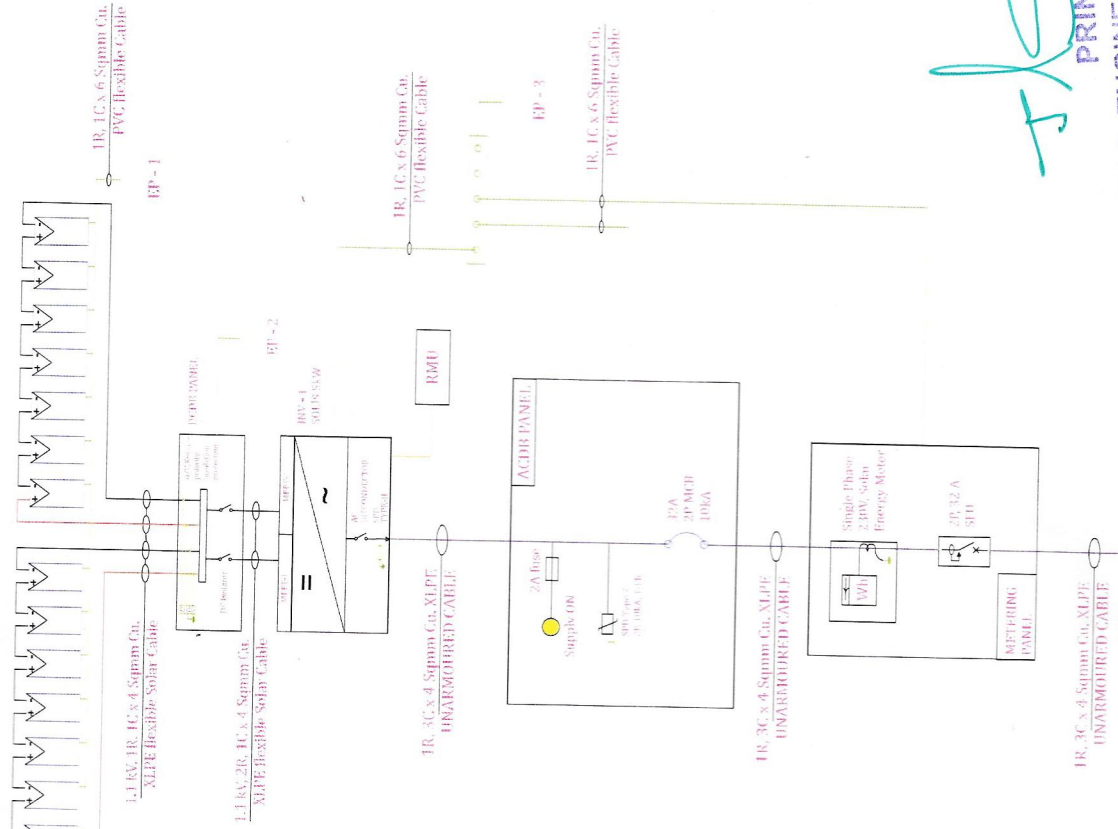
HARDWARE CONNECTION DIAGRAM




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipey,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

SINGLE LINE DIAGRAM OF 5kW ROOFTOP SPV SYSTEM

PV ARRAY RATED CAPACITY (5.025kWp)
15 Nos. 335 Wp DC MODULES



INVERTER DETAILS				
STRING NO.	STRING SIZE	INVERTER NO.	MDPT NO.	CAPACITY (kWp)
01-02	7 x 1.8 x 1	01	01	5.025

PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvolai, Kavaraipettai,
Gummidipondy Taluk,
Thiruvallur Dist - 601 206.

CLIENT MAIN LT PANEL

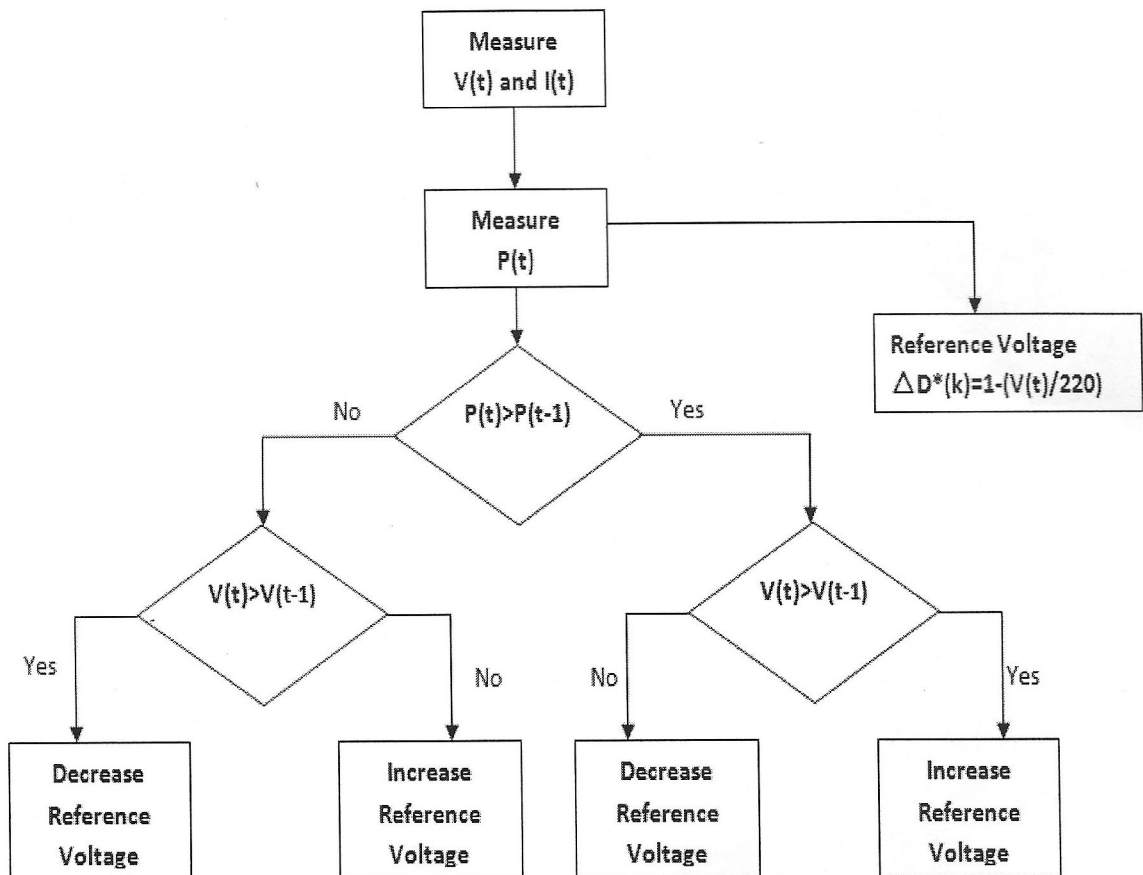
PSEUDO CODE FOR THE PROPOSED MPPT ALGORITHM

- Step 1 : Start the process.
- Step 2 : Initialization of Variables.
- Step 3 : Measure the voltage V and current I from PV array
- Step 4 : Calculate the Power P (i.e., $P=V * I$) from the observed parameters of the panel.
- Step 5 : Calculate the power and voltage deviations ($\Delta p, \Delta V$)
- Step 6 : Multiply both results and assign to a new variable (ΔG).
- Step 7 : Check the variable ($\Delta G > 0$ or $\Delta G < 0$)
- Step 8 : Based on the results of the variable check, increment or decrement the Duty cycle and end the process.



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

FLOW CHART FOR MODIFIED PERTURB AND OBSERVE ALGORITHM



J. J. S.
PRINCIPAL

T.J.S. ENGINEERING COLLEGE
 Peruvoyal, Kavaraipeetai,
 Gummidipoondi Taluk,
 Thiruvallur Dist - 601 206.

POWER EVACUATION

The Direct Current(DC) from modules will be converted into Alternating Current(AC) by Inverters. The inverter outputs are given to a Low Voltage(LT) panels installed inside the School premises.

Monitoring System

Monitoring systems Is mainly used to monitor the performance of the Inverters, energy yield, temperature, irradiance level etc. It provides an extremely flexible interface to facilitate PC-based inverter monitoring via analog modem, GSM, Ethernet, or Internet connections.

Industry-Leading Features and Performance:

- Inverter monitoring parameters include: energy yield, power, array voltages, array currents and AC parameters.
- Environmental-sensor interface capabilities include: temperature, wind speed, and irradiation sensors.

Field-Proven Reliability:

Enhanced remote-monitoring capabilities enable the collection and analysis of critical data needed to facilitate increased power system reliability.

Installer Friendly:

- A front mounted LCD display provides visual confirm at io no fall critical operating parameters.
- Compact easy-to-install package with readily-accessible electrical connections.

Unmatched Applications Flexibility:

- Active alarm management capabilities with automatic delivery of SMS(GSM mode), e-mail, or fax alarm messages.
- Data logger and display-enabled configurations are available.



Jha
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

PROTECTION SYSTEM

Earthing: The array structure of the PV yard will be grounded properly using adequate number of earthing kits. All metal casing / shielding of the plant shall be thoroughly grounded to ensure safety of the power plant.

Lightning Protection: The SPV Power Plant shall be provided with lightning & over voltage protection. The main aim in this protection shall be to reduce the over voltage to a tolerable value before it reaches the PV or other sub system components. The source of over voltage can be lightning, atmosphere disturbances etc. Metal oxide variastor shall be provided inside the Array Junction Boxes. In addition suitable SPD also shall be provided in the Inverter to protect the inverter from overvoltage.



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

DESIGN OF THE POWER PLANT

The Power Plant components are selected & discussed on the previous chapter. Based on the component selection the Power Plant Design is discussed in this chapter.

DESIGN ASSUMPTION FOR CRYSTALLINE MODULES

Location	Chennai
Place of installation	Subam Industries
Latitude	13.0647° N,
Longitude	80.1371° E
Elevation	75 mm
Annual Solar radiation	5.22
Module Facing	SOUTH
Module Tilt angle	15
Sun hours	8.40 – 5.30
Modules in series with the string	As per requirement
String in parallel	As pr requirement
Shading	No shading
NOCT	45
Module efficiency loss	4 %
Power loss at MPP	2 %
Soiling loss	2%

The project is planned to be implemented at the earliest. The most essential aspect regarding the implementation of this project is to ensure that the project is completed within the schedule, spanning from eighteen months from the date of completion of the DPR. A good planning, scheduling, and monitoring program is imperative to complete the project on time and without cost overruns.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 203.

PROJECT IMPLEMENTATION

It is envisaged that the project will have the below mentioned phase of activities. These phases are not mutually exclusive; to implement the project on fast track basis some degree of overlapping is envisaged.

Phase I - Project Development

Phase II - Finalization of the Equipment and accessories

Phase III - Procurement and Construction

Phase IV - Plant Commissioning

Phase I - Project Development

In a power project, development of the project development plays an important role. Almost 50 % of the work is done if one achieves power purchase agreement from the respective state utilities. The project development starts with visits to the region, understanding about the regional conditions, socio economic conditions, transportation facilities and infrastructure facilities available in the region.

Apart from the above the below listed tasks will be under project development:

Preparation of Detailed Project Report (DPR)

Submission of DPR

Power purchase agreement (PPA)

Expedite Central Regulatory Authority clearance

Land acquisition / mortgage

During this phase, a project team will be formed during the execution of the project. The engineers from group will be involved from early stages of execution of the project. This would give them the opportunity to familiarize with the equipment and systems being installed. These personnel should involve with the critical team of installation and commissioning. After the plant being commissioned, these engineers and technicians would occupy key positions in the organization structure for the operation and maintenance of the plant.

The responsibilities of the project team shall be:

Planning and programming of all the resources required for project completion

Inspection of major fabrication items

Organize the construction and commissioning of the plant

Monitoring and controlling the project progress

Execute the project within the planned budget

Phase II- Finalization Of The Equipment And Contracts

In the Power Plant modules and inverters are the lead items and the planning schedule for the project implementation should provide adequate time period for the




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipeetai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

acquisition and installation of these equipments. The specifications for major equipment shall be drawn up at an early stage of the project. Program of design information, from the equipment suppliers, that satisfies the overall project schedule shall be drawn up. Since, the project execution calls for closer coordination among the contractors, consultants and the company, proper contract co-ordination and monitoring procedures shall be made to plan and monitor the project progress.

Phase III - Procurement And Construction

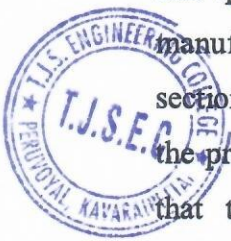
The Procurement is an important function of the implementation of the project. Once the purchase order is placed, the project team follows up regularly to ensure smooth and timely execution of the contract and for obtaining technical information for the inter-package engineering. When the contracts for the equipment are awarded, detailed programme in the form of network are tied up with the supplier to clearly indicate the owner's obligations and the supplier's responsibilities. And upon placement of the purchase order, the project team follows up regularly to ensure smooth and timely execution of the contract and or obtaining technical information for the inter-package engineering. The procurement activity includes review of drawings, expediting, stage and final pre delivery inspection, supervision of installation and commissioning. During Construction, the erection and commissioning phase of all the contracts proceed simultaneously. Adequate power and water shall be made available for the construction. Construction manager takes the overall responsibility of the site.

Phase IV- Erection and Commissioning Phase

The commissioning phase in a project is one where design, manufacturing, erection and quality assurance expertise are put to test. The commissioning team will be from manufacturer of the equipment, consultant and the company. As discussed in the earlier section, staff identified to operate the plant will be involved in the commissioning phase of the project itself. When construction phase is complete, the check list designed to ensure that the plant has been properly installed with appropriate safety measures. The commissioning team will follow the internal operating instructions. The plant shall be subjected to a performance test.

Project Schedule

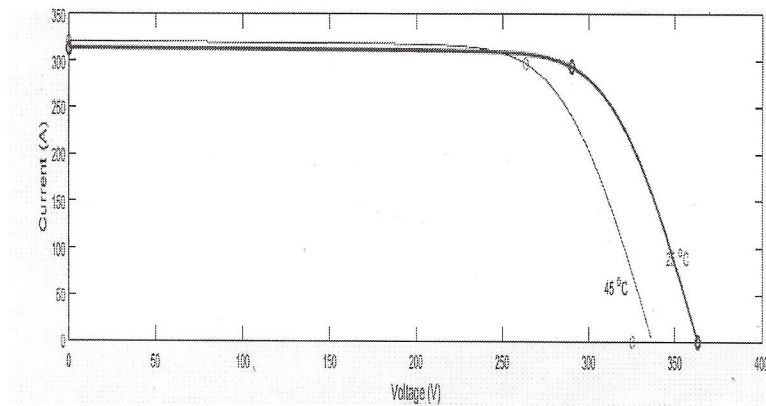
The overall Project completion is scheduled for 9 months from the date of finalization of contract. The same can be discussed and agreed mutually during the detail discussion. Based on the design and technical discussion from the above Chapters the schedule and the Bill of Material for both the option is provided below



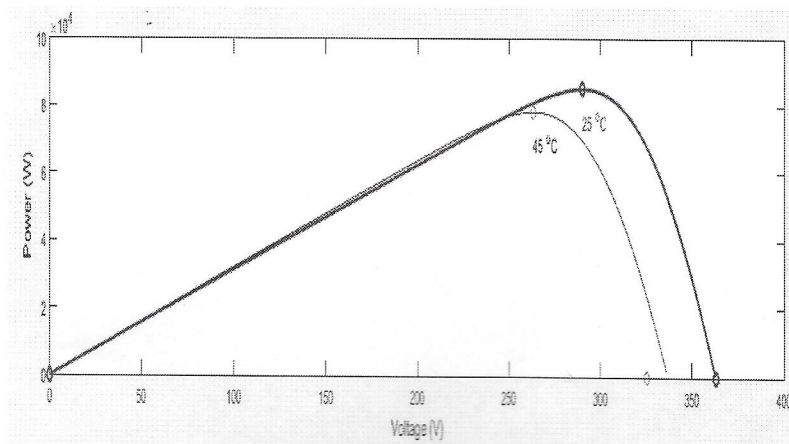
Handwritten signature in green ink.
PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvolai, Kanniampettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

SIMULATION RESULTS

I-V CURVE

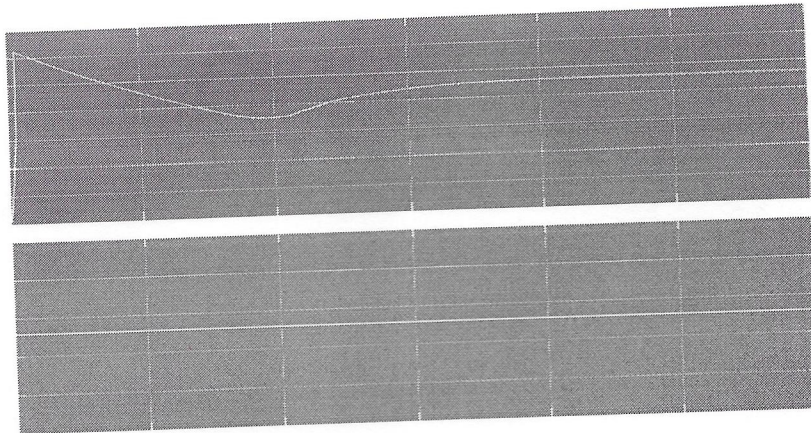


P-V CURVE




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipttai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

CONVERTER OUT PUT VOLTAGE



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

PROJECT IMPLEMENTATION



PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Conclusion

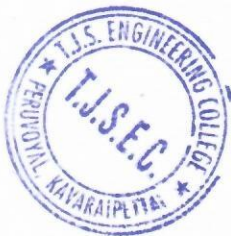
The implementation of solar photovoltaic (PV) systems has proven to be a significant step towards sustainable energy generation and environmental conservation. Through this report, we have examined the various aspects of 5 kV solar PV implementation, including technological advancements, economic considerations, and environmental benefits. Solar PV systems offer a reliable and cost-effective solution for electricity generation, reducing reliance on fossil fuels and mitigating greenhouse gas emissions. The declining costs of solar PV components, coupled with supportive policies and incentives, have facilitated widespread adoption across residential, commercial, and industrial sectors. Furthermore, the implementation of PV panel has demonstrated significant socio-economic benefits, including job creation, energy independence, and enhanced energy security. Communities and businesses embracing solar energy have witnessed reduced energy bills and increased resilience to power disruptions. However, challenges such as intermittency, grid integration, and policy uncertainties remain, necessitating continuous innovation and strategic planning. Addressing these challenges will require collaborative efforts from governments, industry stakeholders, and research institutions.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.

Reference

1. Soewarto Tossin and A Andy BW, "Load Optimization on 2000Wp OFF-GRID Solar Power Plant" Optimaliasi Beban pada Pembangkit Listrik Tenaga Surya OFF-GRID 2000Wp", *Prosiding Seminar Nasional Teknik Elektro*, vol. 5, pp. 298-300, 2020.
2. M Nuruddin, IW Riski, F Ahmad and A. Saiful, "Design Of The Off-Grid Solar Power Plant Installation System For Cooking Process Using Electric Stove", *The Second International Conference on Food and Agriculture*, pp. 578-583, 2019.
3. Jaka Windarta, Enda Wista Sinuraya, Zaenal A Ali, S Andalas Era and K Angghika, "Application of Solar Power Generation Technology (PLTS) at SMA Negeri 6 Surakarta as an Energy Saving and Environmentally Friendly School "Penerapan Teknologi Pembangkit Listrik Tenaga Surya (PLTS) di SMA Negeri 6 Surakarta sebagai Sekolah Hemat Energi dan Ramah Lingkungan", *Pro siding Seminar Nasional MIPA UNIBA*, 2019.
4. Al-S Ali Najah, Mohd Y Hj Othman, K. Sopian, Sohif Mat, M.H. Ruslan and Azher M. Abed, "Design & Sizing of Stand-alone Solar Power Systems A house Iraq", *Conference: Recent Advances in Renewable Energy Sources*, pp. 145-150, April 2015.
5. Muhammad Fahmi Hakim, "Design of Rooftop Off Grid Solar Panels in Residential Homes as an Alternative Source of Electrical Energy "Perancangan Rooftop Off Grid Solar Panel Pada Rumah Tinggal Sebagai Alternatif Sumber Eenergi Listrik", *Jurnal Dinamika DotCom*, vol. 8, no. 1, Januari 2017.




PRINCIPAL
T.J.S. ENGINEERING COLLEGE
Peruvoyal, Kavaraipettai,
Gummidipoondi Taluk,
Thiruvallur Dist - 601 206.